

Elective course (Term 7)- Sept 2025

01.119/99.503: Nanoelectronics and Technology

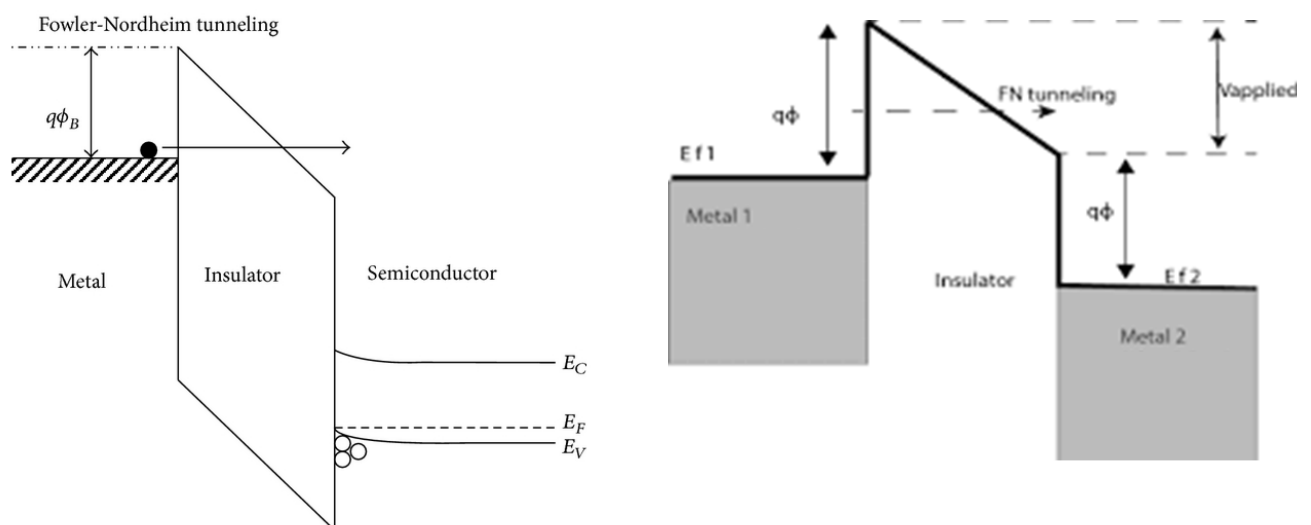
Extra Problems with solutions

1. Explain F-N tunnelling, direct tunnelling and trap-assisted tunnelling in gate dielectrics with the help of energy band diagrams. Highlight the significance of these tunnelling phenomena on dielectric reliability and failure.

ANS:

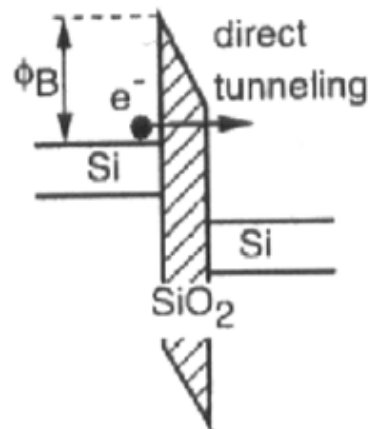
F-N tunnelling: Fowler–Nordheim tunneling is the wave-mechanical tunneling of an electron through a rounded triangular barrier by applying a high electric field. F-N tunneling is relevant mechanism under higher applied voltage/electric field.

At Fowler-Nordheim conditions the electron tunnels through a part of the barrier to the conduction band of the insulator. From there it can flow to the anode. Most likely to dominate current in MOS structures with oxide thickness more than 4-5 nm.



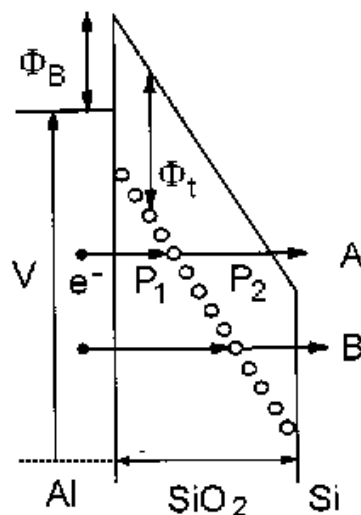
Direct tunnelling: At direct tunneling conditions the electron tunnels through the whole. energy barrier.

Direct tunneling in MOS structures with ultra-thin oxide (< 4 nm) during which electrons from the conduction band in semiconductor are transferred across the oxide directly (i.e. without changing energy) into the conduction band of metal. Probability of direct tunneling is a very strong function of the width of the barrier electron tunnels through (oxide thickness in MOS devices). Direct tunneling can occur even at low applied voltage.



Trap-assisted tunnelling:

The presence of traps/defects inside the gate oxide, splits the energy barrier in different parts, thus allowing the consecutive tunnel through thinner energy barriers and increasing the probability of the total tunnelling process. When the tunneling mechanism is strongly dependent on the oxide thickness and the oxide traps become most favorable condition for the TAT process and largely increasing the tunneling current. Therefore, the stress-induced distortion of the tunneling characteristic is more enhanced in the DT regime, hence at low gate voltages.



(Refer to study material for details on different tunneling mechanisms.)

2. Name the different dielectric degradation mechanisms/models and briefly discuss.

ANS:

Degradation and Breakdown mechanism:

- Defect generation (oxygen vacancy formation)- SILC and TDDB
- Percolation path formation-Soft BD
- Thermal runaway- Hard BD, catastrophic failure

The dielectric degradation mechanisms can be explained using the following models

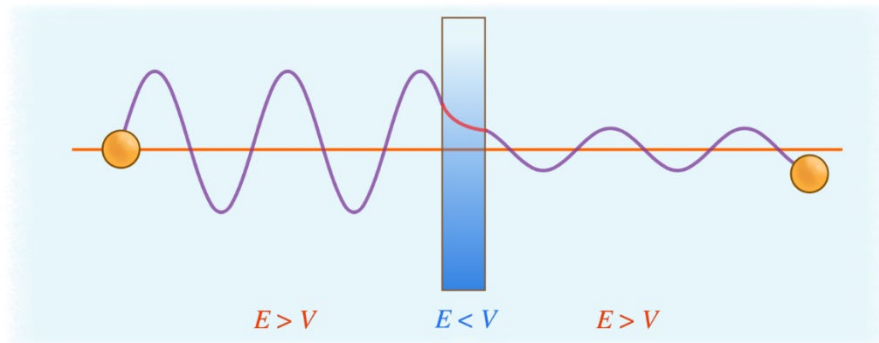
- Hydrogen release model
- Field driven thermo-chemical model
- Anode-hole injection model

(Refer to study material for details on dielectric degradation and breakdown. Also for details on Hydrogen release model, thermos-chemical model and Anode-Hole injection model)

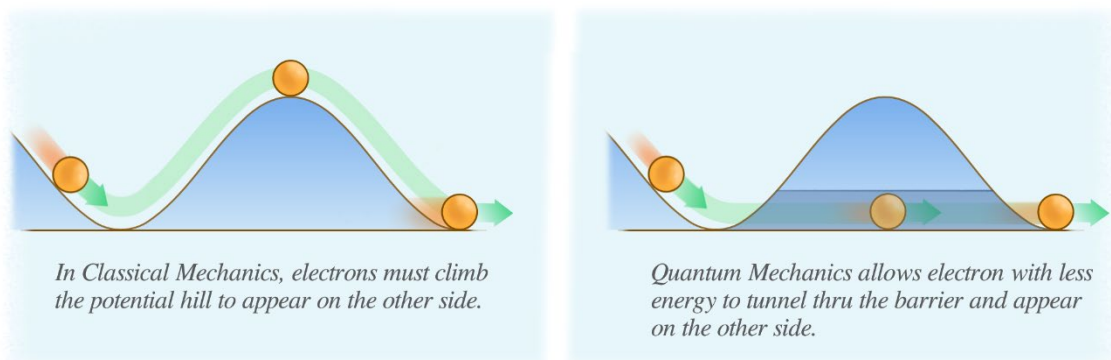
3. Compare quantum mechanical tunnelling of charges in thin dielectric with classical mechanics with the help of figures. Highlight the key nanoscale characterization technique working on the basis of quantum mechanical tunneling effect.

Ans: Quantum tunneling is a phenomenon in which particles penetrate a potential energy barrier with a height greater than the total energy of the particles.

The phenomenon is interesting and important because it violates the principles of classical mechanics.



In classical mechanics, if the ball does not have enough kinetic energy ($E < V$), to overcome the barrier it will never roll from one valley to the other. In contrast, when quantum effects are taken into effect, the ball can "tunnel" through the barrier to the other valley, even if its kinetic energy is less than the potential energy of the barrier to the top of one of the hills.



For a quantum particle to *appreciably* tunnel through a barrier three conditions must be met:

- The height of the barrier must be finite and the thickness of the barrier should be thin.

- The potential energy of the barrier exceeds the kinetic energy of the particle ($E < V$).
- The particle has wave properties because the wave-function is able to penetrate through the barrier. This suggests that quantum tunneling only apply to microscopic objects such protons or electrons and does not apply to macroscopic objects.

This amazing property of microscopic particles play important roles in explaining several physical phenomena including the principle of tunneling leading to the development of Scanning Tunneling Microscope (STM) which had a profound impact on material science research.

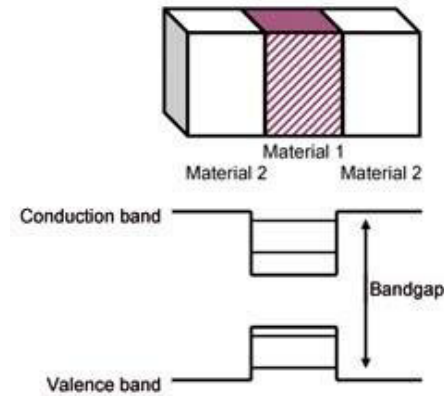
In STM, the quantum mechanical tunnelling occurs between the STM tip and sample surface through the vacuum.

4. Explain the discrete energy level formation in quantum well (2D) with 1D carrier confinement with the help of energy band diagram. Also, describe its impact on charge carrier transport.

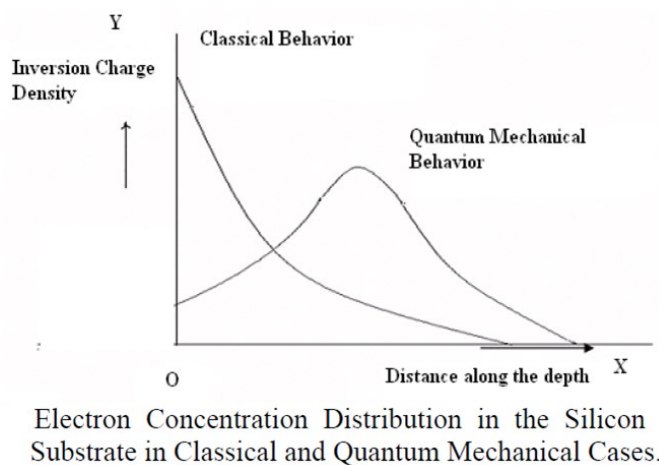
Ans: The effects of quantum confinement take place when the quantum well thickness becomes comparable to the de Broglie wavelength of the carriers (generally electrons and holes), leading to energy levels called "energy sub-bands", i.e., the carriers can only have discrete energy values.

As a result, the energy difference between energy bands is increased with decreasing in particle size dimension. Particle behaves like a free particle when the dimensions of the confining structure are very large in comparison to the de Broglie wavelength. On this stage, the energy states are continuous and the bandgap is coming to its original position and another energy spectrum does not remain continuous and becomes discrete in nature when the dimensions of confining structure are decreased towards nanoscale. Therefore, the bandgap exhibits size dependence properties, and eventually causes a blue shift in the emitted light as the

particle's size is decreased. However, this effect demonstrates the consequences of confining the electrons and charge carriers are not free to move in all directions.



5. The electron concentration at the SiO_2 -Silicon interface (in Silicon substrate side) in classical and quantum mechanical conditions is shown as below.



Discuss the reason for the shift in the peak value of electron concentration of charges away from the interface (towards the silicon substrate) under quantum mechanical conditions. Use energy band diagram of SiO_2 -Silicon interface to explain this phenomenon.

Ans:

Energy quantization occurs in the channel near the interface of the oxide layer and the silicon channel because of the presence of the strong electric field. As illustrated by Figure (a), quantization leads to the splitting of the continuous energy band and the formation of sub-bands with a two-dimensional (2-D) density of states. Because of the smaller density of states in the 2-D system, the net sheet charge density of carriers in the channel is lower than that calculated from the classical (3-D) case, as shown in Figure above (question part). Thus, it requires a larger gate voltage to generate the same charge sheet density in the 2-D inversion layer as that in the corresponding 3-D case. Consequently, the threshold voltage increases when energy quantization is considered.

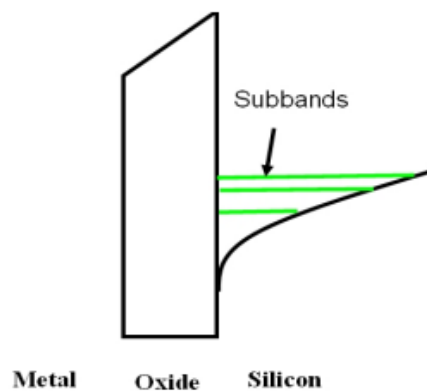


Figure (a)
Discrete energy levels due to quantization.

The results show the two distinctive differences of between the quantum-mechanical solution and the classical models: (1) the channel carriers are distributed among discrete energy levels with 2-D density of states instead of the single continuous energy band with 3-D density of states; (2) the peak of the space carrier concentration is located some distance away from the surface of the substrate, which leads to a finite thickness of inversion layer.